A Map of Vibe Space: Navigating the Web by Aesthetic Character

Ari Holtzman and NeuriCo

Abstract

Current web navigation organizes content by topic, authority, and behavioral signals, but no existing system captures the *aesthetic character*—or *vibe*—of a web page. We propose a pipeline that prompts a large language model with web page text and the question "What is this web page giving?" to elicit concise vibe descriptions in contemporary internet vernacular. We embed these descriptions with a sentence transformer and project them to two dimensions with UMAP, producing a navigable *vibe map* in which proximity reflects aesthetic similarity rather than topical overlap. On 497 pages sampled from the C4 corpus, we find that vibe space organizes content in a fundamentally different way from raw-text embedding space: the Adjusted Rand Index between the two clusterings is only 0.113 (p=0.0002), and vibe-space nearest neighbors have dramatically lower content similarity than raw-space neighbors (Cohen's d=1.44). We identify 293 cross-domain page pairs—pages about entirely different subjects that cluster together because they share the same aesthetic energy. These results establish that LLM-generated vibe descriptions capture a genuinely novel dimension of web content, opening the door to *browse-by-vibe* as a new modality for web discovery.

1 Introduction

How people find web content has changed remarkably little since the advent of search engines. Users type keywords, follow hyperlinks, or accept algorithmic recommendations—all of which organize the web by *topic*, authority, or past behavior. Yet when someone describes a web page informally, they rarely lead with its subject matter. Instead, they reach for aesthetic shorthand: a site "gives early-2010s personal blog energy," or "has that corporate LinkedIn vibe." This subjective, cultural dimension—what we call the *vibe* of a page—is invisible to every existing navigation system.

Why does vibe matter? Vibes capture information orthogonal to topic: a Romanian collaboration blog and a British movie review site may share nothing in subject matter, yet both project the same warm, hand-crafted, personal-blog energy. A user looking for content that *feels* a certain way has no tool to find it; current systems cannot surface the latent aesthetic structure of the web. Recent work has begun to formalize the notion of "vibes" as measurable axes of variation in LLM outputs Dunlap et al. [2025], but this line of research has not yet been applied to web content at large.

Our approach. We propose a simple three-stage pipeline (figure 1): (1) prompt a large language model with the text of a web page and ask "*What is this web page giving? Start your answer with 'it's giving'*," (2) embed the resulting vibe description with a sentence transformer, and (3) project the embeddings to two dimensions with UMAP to produce an interactive *vibe map*. We compare this vibe space against a baseline content space formed by directly embedding the raw page text. On 497 pages from the C4 corpus Raffel et al. [2020], vibe space and content space disagree substantially: the Adjusted Rand Index between their clusterings is only 0.113 (p=0.0002), and vibe-space neighbors exhibit a Cohen's d=1.44 reduction in raw-text similarity relative to content-space neighbors. We find 293 cross-domain page pairs that cluster together in vibe space despite covering entirely different topics.

Our contributions are:

- We introduce a pipeline that transforms web pages into vibe descriptions using a culturally grounded LLM prompt and show that the resulting embeddings capture aesthetic similarity distinct from topical similarity.
- We provide quantitative evidence that vibe space organizes pages differently from content space, with strong statistical significance and large effect sizes across multiple evaluation metrics.
- We demonstrate cross-domain discovery: pages about unrelated subjects cluster together when they share aesthetic character, enabling a fundamentally new mode of web browsing.

The remainder of this paper is organized as follows. Section 2 reviews related work on embedding visualization, LLM-based content description, and style-aware representations. Section 3 describes our pipeline and evaluation methodology. Section 4 presents quantitative and qualitative results, and section 5 discusses limitations and future directions.

2 Related Work

Formalizing vibes in NLP. Dunlap et al. [2025] introduced VIBECHECK, a system that discovers and quantifies qualitative "vibe" axes—tone, formatting, writing style—along which LLM outputs differ. Their pipeline uses an LLM to examine output pairs, discover vibe axes (e.g., "formal $\rightarrow$ friendly"), and score outputs on each axis. While VIBECHECK demonstrates that vibes are measurable and predictive (achieving 80% accuracy at identifying model identity), it is designed for comparing LLM outputs, not for characterizing web content or constructing navigable embedding spaces. We extend the concept of vibes from model comparison to web-page description and embed the resulting descriptions to create a spatial map.

Embedding visualization. Interactive visualization of high-dimensional embeddings is a well-studied problem. Smilkov et al. [2016] introduced the Embedding Projector for exploring word and image embeddings via t-SNE, UMAP, and PCA. Cheng et al. [2023] proposed WIZMAP, which scales to millions of points using multi-resolution map-like navigation. Most recently, Ren et al. [2025] presented Embedding Atlas, a browser-based tool that supports real-time UMAP computation, density contours, and cross-filtering on millions of data points. All of these tools visualize *content* or *semantic* embeddings; none address the question of whether a different embedding—one capturing aesthetic character rather than topic—would produce a qualitatively different and useful map. Our work contributes a new kind of input to these visualization tools: vibe embeddings.

Interpreting embedding spaces. Tennenholtz et al. [2024] proposed ELM (Embedding Language Model), which trains adapter layers that allow an LLM to generate natural-language descriptions of entities from their embeddings. Liu et al. [2024] developed gradient-based methods to overlay spatial word clouds on 2D document projections, making regions of embedding space interpretable. Haschka et al. [2025] build hierarchical semantic trees from embedding spaces, revealing multi-scale cluster structure. These approaches interpret *existing* embedding spaces; we construct a *new* embedding space by first translating content into vibe descriptions and then embedding those descriptions.

Style and aesthetic representation. Recent work has examined how writing style, as distinct from topic, shapes the geometry of embedding spaces Authors [2025]. Authors [2026] model human concept navigation as trajectories through transformer embedding spaces, providing a theoretical framework for browsing semantic maps. Our work is complementary: we show that vibe descriptions—which capture style, era, community type, and cultural energy—produce embeddings that organize content along aesthetic axes rather than topical ones.

LLMs and informal language. The "it's giving" prompt we use draws on contemporary internet slang. Authors [2024] and Wilson et al. [2024] study how LLMs comprehend and produce slang, finding that models trained on large web corpora handle informal language reasonably well. Gur et al. [2022] demonstrate that LLMs pretrained on natural language transfer effectively to HTML understanding tasks. We build on these findings by prompting an LLM with web page text and a slang-inflected question, relying on the model's cultural fluency to produce evocative, aesthetic descriptions.

3 Methodology

We describe a three-stage pipeline that transforms web page text into a two-dimensional vibe map and detail the evaluation protocol we use to assess whether vibe space captures information distinct from content space.

3.1 Data

We draw from the C4 (Colossal Clean Crawled Corpus) small sample Raffel et al. [2020], which contains 10,000 pre-extracted web page texts hosted on HuggingFace. After filtering to pages with at least 200 characters (removing 447 pages), we randomly sample 500 pages with seed 42. Three pages fail during LLM processing due to transient API errors, yielding a final working sample of $N{=}497$.

3.2 Vibe Generation

For each page, we truncate the text to 2,000 characters and prompt GPT-4.1 with the following:

> **System:** "You are a cultural commentator who describes the vibe, aesthetic, and energy of things using contemporary internet language. Be specific, evocative, and concise (1–3 sentences)."
>
> **User:** "Here is the content of a web page: [TEXT]. What is this web page giving? Start your answer with 'it's giving'."

We use temperature 0.7 and a maximum of 150 tokens. The prompt is designed to elicit descriptions that foreground aesthetic character—era, tone, community type, cultural energy—rather than topical content. The "it's giving" framing draws on Gen Z internet vernacular, encouraging the model to adopt a culturally fluent register.

3.3 Embedding and Projection

We embed both the raw page texts and the vibe descriptions using ALL-MPNET-BASE-V2 Reimers and Gurevych [2019], a 768-dimensional sentence transformer. We then project both sets of embeddings to two dimensions using UMAP McInnes et al. [2018] with $n_{\text{neighbors}}{=}15$, min_dist$=0.1$, and cosine distance—standard settings that balance local and global structure preservation.

3.4 Evaluation Protocol

We evaluate four hypotheses about the relationship between vibe space and content space.

H1: Vibe diversity. We measure whether the LLM produces meaningfully varied descriptions by computing the number of unique descriptions, vocabulary richness (type-token ratio), and mean description length.

H2: Vibe $\neq$ topic. We cluster both embedding spaces with KMeans Arthur and Vassilvitskii [2007] for $k \in \{5, 10, 15, 20\}$ and compute the Adjusted Rand Index (ARI) Hubert and Arabie [1985] between the two clusterings. Low ARI indicates that vibe space groups pages differently from content space. We assess significance with a permutation test (10,000 permutations).

H3: Cluster quality. We compute silhouette scores Rousseeuw [1987] for each space and compare them using bootstrap 95% confidence intervals (1,000 resamples).

H4: Cross-domain discovery. For each page, we identify its $K{=}5$ nearest neighbors in both vibe space and content space (using high-dimensional cosine similarity). We measure (a) the overlap between the two neighbor sets, and (b) the raw-text cosine similarity of vibe-space neighbors. If vibe space enables cross-domain discovery, vibe-space neighbors should have substantially lower content similarity than content-space neighbors. We report the Mann–Whitney U statistic and Cohen's d effect size.

Table 1: Clustering comparison between vibe space and content space across values of k. Silhouette scores measure within-space cluster quality Rousseeuw [1987]; ARI measures agreement between the two clusterings Hubert and Arabie [1985]. Vibe space consistently achieves higher silhouette scores and low agreement with content space. Best silhouette per row in **bold**.

k	Silhouette (Content)	Silhouette (Vibe)	ARI
5	0.033	**0.064**	0.137
10	0.037	**0.053**	0.113
15	0.035	**0.060**	0.103
20	0.037	**0.059**	0.100

4 Results

4.1 Vibe Descriptions Are Maximally Diverse (H1)

GPT-4.1 produces a unique vibe description for every one of the 497 pages. Across 18,462 total words, we observe 6,217 unique word types, yielding a type-token ratio of 0.337. The mean description length is 254 characters ($\pm$61). Of the 497 descriptions, 493 (99.2%) contain the phrase "it's giving," with 33 using the exact lowercase form and the remainder using capitalized or minor variants. The model consistently employs culturally fluent internet vernacular—terms like "cottagecore," "main character energy," and various "-core" suffixes appear naturally throughout the descriptions.

4.2 Vibe Space Captures Different Information Than Content Space (H2)

table 1 presents clustering comparison results across multiple values of k. The ARI between vibe-space and content-space clusterings ranges from 0.100 to 0.137, indicating that only 10–14% of page-pair co-cluster assignments agree. A permutation test at k=10 yields p=0.0002, confirming that the observed agreement, while statistically above chance, is very low—vibe space and content space organize pages in fundamentally different ways.

4.3 Vibe Space Has Better Cluster Structure (H3)

Across all values of k, vibe-space silhouette scores (0.053–0.064) exceed content-space scores (0.033–0.037). Bootstrap 95% confidence intervals at k=10 are $[0.056, 0.080]$ for vibe space and $[0.046, 0.061]$ for content space. While both spaces have modest absolute silhouette scores—expected given the diversity of web content—vibe descriptions produce more coherent groupings than raw text.

4.4 Vibe Space Enables Cross-Domain Discovery (H4)

table 2 summarizes the nearest-neighbor analysis. Pages share only 15.1% of their K=5 neighbors between the two spaces, compared to 1.0% expected by chance. More strikingly, the raw-text cosine similarity of vibe-space neighbors (0.178 ± 0.132) is dramatically lower than that of content-space neighbors (0.339 ± 0.087). This difference is highly significant (Mann–Whitney U, $p < 10^{-6}$) with a very large effect size (Cohen's d=1.44).

These results confirm that vibe space brings together pages from different content domains that share aesthetic character. We identify 293 cross-domain page pairs—pages whose vibe-space neighbors have raw-text cosine similarity below 0.15, indicating topically unrelated content.

4.5 Qualitative Analysis

Figure 1 shows the raw-text and vibe-space UMAP projections side by side. The vibe map reveals coherent aesthetic regions that cut across topics. We highlight three illustrative cross-domain clusters:

Table 2: Nearest-neighbor analysis (K=5). Vibe-space neighbors have substantially lower content similarity than content-space neighbors, confirming that vibe space brings together pages from different content domains.

Metric	Value
Mean neighbor overlap	0.753 / 5 (15.1%)
Expected random overlap	0.050 / 5 (1.0%)
Content sim. of content-space neighbors	0.339 ± 0.087
Content sim. of vibe-space neighbors	0.178 ± 0.132
Mann–Whitney U p-value	$< 10^{-6}$
Cohen's d	**1.44**

Figure 1: UMAP projections of 497 web pages in content space *(Left)*and vibe space *(Right)*. Points are colored by content-space cluster assignment in both panels. The rearrangement of colors in the right panel confirms that vibe space organizes pages differently from content space.

Early-2010s personal blog energy. A Romanian collaboration blog and a British movie review site—different languages, different subjects—cluster together because both project the same warm, hand-crafted, personal-blog aesthetic.

Cozy niche forum energy. Gaming forums, tech support forums, and train simulator forums share "cozy niche forum energy" despite wildly different content. The vibe captures the community format, conversational tone, and retro web design rather than the subject matter.

Corporate LinkedIn energy. Pages about corporate acquisitions, academic engineering programs, and professional development services cluster together, unified by polished, institutional communication style.

4.6 Dimensionality Analysis

Vibe embeddings require 128 principal components to explain 90% of variance, compared to 178 for raw-text embeddings. Vibe space is more compact, suggesting that aesthetic character occupies a lower-dimensional manifold than topical content—there are fewer "vibe dimensions" than "topic dimensions."

5 Discussion

Vibe is a real and distinct signal. Our results provide strong evidence that LLM-generated vibe descriptions capture information genuinely different from what raw-text embeddings encode. The low ARI (0.10–0.14), large effect size on neighbor content similarity (Cohen's d=1.44), and consistent results across all values of k all point to the same conclusion: vibe space is not a noisy version of content space but a qualitatively different organization of web content.

Why does the pipeline work? We hypothesize that the "it's giving" prompt activates a specific mode of cultural commentary in the LLM—one trained on internet discourse where aesthetic judgments are common. The LLM compresses each page's content, layout cues (reflected in text structure), and cultural markers into a concise aesthetic summary. The sentence transformer then embeds these summaries in a space where aesthetic similarity translates to geometric proximity.

Implications for web navigation. The existence of a coherent vibe space suggests that "browse by vibe" is a feasible navigation modality. A user could select a page they enjoy and ask for pages that "give the same energy," surfacing content from entirely different domains. This is complementary to topic-based search: where search answers "what is this about?," vibe navigation answers "what does this *feel* like?"

Limitations. Several limitations qualify our findings. First, our sample of 497 pages, while sufficient for statistical testing, is far from web scale. Whether vibe clusters remain coherent at millions of pages is an open question. Second, we use pre-extracted text from C4 rather than live HTML; visual design elements (colors, fonts, layout) contribute to vibe but are not captured by our text-only pipeline. Third, we test only one LLM (GPT-4.1) with one prompt; different models or prompts may produce different vibe characterizations. Fourth, we lack a user study: we demonstrate structural properties of vibe space but do not test whether humans find it useful for browsing. Fifth, the C4 corpus is predominantly English, and vibe characterizations may be culturally specific to English-language internet culture. Finally, cost is a barrier to scale: at approximately \$0.01 per page, processing one billion pages would cost roughly \$10 million without optimization.

Broader impact. A vibe-based navigation system could help users discover content outside their usual information bubbles—not by recommending opposing viewpoints (as diversity interventions typically do), but by connecting content that shares aesthetic character across domain boundaries. However, vibes are subjective and culturally situated. A system that foregrounds "vibes" risks encoding the aesthetic preferences of the LLM's training data, which skews toward English-language, Western internet culture. Deploying such a system responsibly would require careful attention to whose vibes are represented and whose are marginalized.

6 Conclusion

We have shown that prompting an LLM with web page text and asking "What is this web page giving?" produces vibe descriptions that, when embedded, create a space fundamentally different from raw-text embedding space. Vibe space captures aesthetic similarity—era, tone, community type, cultural energy—rather than topical similarity, with strong statistical support: ARI of 0.113 (p=0.0002) between clusterings and a Cohen's d of 1.44 on cross-space neighbor content similarity. We identify 293 cross-domain page pairs that cluster together by shared aesthetic character despite covering unrelated subjects.

These results establish the feasibility of *browse-by-vibe* as a new modality for web discovery. Future work should scale the pipeline to larger corpora, incorporate visual features from page screenshots, conduct user studies to evaluate browsing utility, and compare vibe descriptions across multiple LLMs and prompt formulations. We are particularly interested in whether a small, fine-tuned model can replicate the vibe-description capability of GPT-4.1 at a fraction of the cost, which would be necessary for web-scale deployment.

References

David Arthur and Sergei Vassilvitskii. k-means++: The advantages of careful seeding. In *Proceedings of the ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2007.

Various Authors. SLANG: New concept comprehension of large language models. *arXiv preprint arXiv:2401.12585*, 2024.

Various Authors. Embedding style beyond topics. *arXiv preprint arXiv:2501.00828*, 2025.

Various Authors. Characterizing human semantic navigation as trajectories in embedding space. *arXiv preprint arXiv:2602.05971*, 2026.

Zijie J Cheng, Felipe Maia Polo, et al. WizMap: Scalable interactive visualization for exploring large machine learning embeddings. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

Lisa Dunlap, Krishnakant Mandal, Trevor Darrell, Jacob Steinhardt, and Joseph E Gonzalez. VibeCheck: Discover and quantify qualitative differences in large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.

Izzeddin Gur, Ofir Nachum, Yingjie Miao, Mustafa Safdari, Austin Huang, Aakanksha Sharma, Saurabh Gupta, Shrestha Shah, and Aleksandra Faust. Understanding HTML with large language models. *arXiv preprint arXiv:2210.03945*, 2022.

Thomas Haschka et al. Discovering multi-scale semantic structure in text corpora. *arXiv preprint arXiv:2512.23471*, 2025.

Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985.

Guo Liu, Chris North, and Aleksandra Faust. Visualizing spatial semantics of dimensionally reduced text embeddings. *arXiv preprint arXiv:2409.03949*, 2024.

Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

Donghao Ren, Fred Hohman, Tiffany Lin, and Dominik Moritz. Embedding Atlas: Low-friction, interactive embedding visualization. *arXiv preprint arXiv:2505.06386*, 2025.

Peter J Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.

Daniel Smilkov, Nikhil Thorat, Charles Nicholson, Emily Reif, Fernanda B Viégas, and Martin Wattenberg. Embedding projector: Interactive visualization and interpretation of embeddings. *arXiv preprint arXiv:1611.05469*, 2016.

Guy Tennenholtz, Yinlam Chow, Chih-Wei Hsu, Jihwan Jeong, Lior Shani, and Craig Boutilier. Demystifying embedding spaces using large language models. *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

Zhewei Wilson et al. Toward informal language processing: Knowledge of slang in large language models. *arXiv preprint arXiv:2404.02323*, 2024.